

Mixed-Type Wafer Map Dataset
52x52 images, 38 defect patterns



Preprocessing
One-hot encode pixels →
52x52x3



Train / Val / Test Split
70% / 15% / 15%
Iterative stratification



Data Augmentation
Rotations, flips, shifts
Training only



Focal Loss + Class Weights
Handle class imbalance



Phase 1: Multi-Label
8 base defect types



Custom CNN



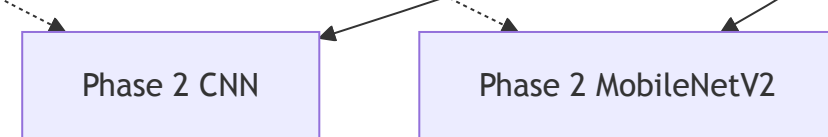
MobileNetV2
Transfer Learning

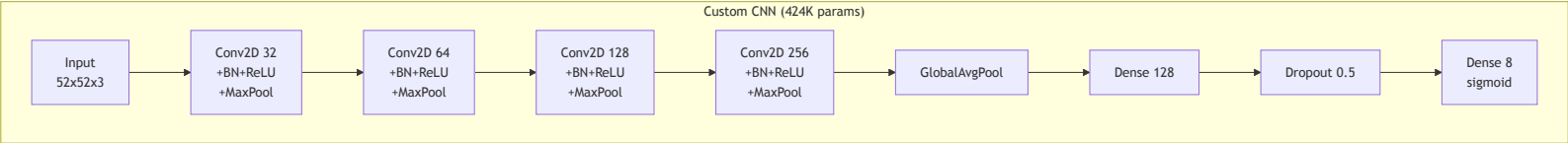


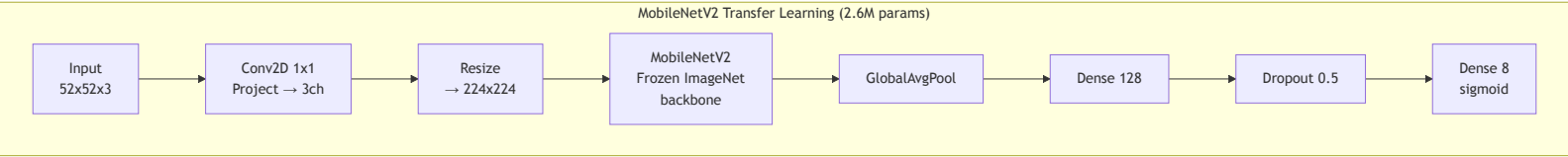
Phase 2: Multi-Class
38 defect patterns

Freeze backbone
Replace head

Freeze backbone
Replace head







Phase 2 Model (both variants)

